Technical Documentation: Project Pathfinder: New Home

Level Types and Progression System

Level progression system

Project Pathfinder: New Home employs a level progression system inspired by a reversed tree structure. Each galaxy in the game is represented as a reversed tree structure with levels as nodes. The tree is visually presented in the UI, allowing players to navigate and make strategic decisions on their path through the galaxy.

Players start at the base of the tree (the beginning of the galaxy). As they complete each level, they ascend toward the single end-point: the Boss level. The reversed tree structure for each galaxy is randomly generated, providing variety and unpredictability in each run. Occasionally, the structure may present side branches that connect different paths, providing players with the opportunity to switch their chosen path mid-run.

Completing a Boss level rewards the player with progression to the next galaxy, where a new reversed tree is generated.

1. Shop Level

The Shop level allows players to use their collected Scraps to repair their ship, purchase new equipment, and upgrade existing ones. The shop level always comes before the Boss level.

- Key Components:

- Equipment Catalog: A list of available equipment for purchase or upgrade.
- Scraps Currency System: An in-game currency system using Scraps.

- Flow:

- Player enters the shop level.
- Player interacts with the shop, browsing the available equipment.
- Player uses collected Scraps to purchase or upgrade equipment.
- After finishing their transactions, the player exits the shop and proceeds to the next level.

2. Persistent Currency Reward Level

The Persistent Currency Reward level provides players with the opportunity to collect a larger quantity of the game's persistent currency, which can be used outside of the run for various upgrades.

- Key Components:

- Currency Nodes: Nodes scattered throughout the level that the player can collect to gain persistent currency.

- Flow:

- Player enters the persistent currency reward level.
- Player navigates the level, collecting currency nodes.
- After collecting the nodes, the player exits the level and proceeds to the next one.

3. Scraps Reward Level

In the Scraps Reward level, players encounter a larger number of enemies that drop a high amount of Scraps, which can be used in the Shop level. At the end of the level, players are rewarded with a bonus amount of Scraps.

- Key Components:

- Enemy Encounters: Numerous enemy encounters that drop Scraps when defeated.

- End-Level Bonus: A bonus amount of Scraps rewarded at the end of the level.

- Flow:

- Player enters the Scraps reward level.
- Player engages in combat with enemies, collecting Scraps upon their defeat.
- After defeating all enemies and reaching the end of the level, player is rewarded with a bonus amount of Scraps.
- Player exits the level and proceeds to the next one.

4. Equipment Reward Level

The Equipment Reward level provides players with the opportunity to earn new equipment for their ship. At the end of the level, players are presented with a choice of three random pieces of equipment.

- Key Components:

- Equipment Drops: Random equipment drops that players can collect.
- End-Level Choice: A choice between three pieces of equipment rewarded at the end of the level.

- Flow:

- Player enters the equipment reward level.
- Player navigates the level, collecting equipment drops.
- Upon reaching the end of the level, player is presented with a choice of three pieces of equipment.
- Player chooses one piece of equipment to take, then exits

the level and proceeds to the next one.

5. Repair Level

The Repair level is a level without enemies, where the player can repair their ship. It occurs in the middle or toward the end of the tree, but never on the first two branches. A galaxy may have no more than one Repair level and there is a chance to have none at all.

- Key Components:

- Repair Station: A station where the player can repair their ship.

- Flow:

- Player enters the repair level.
- Player interacts with the repair station to repair their ship.
- After finishing repairs, the player exits the level and proceeds to the next one.

6. Mystery Level

The Mystery level is a unique level that can take the form of any other level type, except the Shop level. The rewards in a Mystery level are greater than in normal levels of the same type.

- Key Components:

- Random Level Type: The Mystery level randomly assumes the form of any other level type, except the Shop level.
- Increased Rewards: Rewards in the Mystery level are greater than in normal levels of the same type.

- Flow:

- Player enters the Mystery level.
- The level assumes the form of a random level type, except the Shop level.
- Player navigates the level according to its assumed form and collects the increased rewards.
- Player exits the level and proceeds to the next one.

7. Boss Level

The Boss level is the final level in each galaxy. Upon completion, the player progresses to the next galaxy. Defeating the boss yields a special Boss currency that can be used outside of the run for special upgrades.

- Key Components:

- Boss Enemy: A challenging enemy that the player must defeat to complete the level.
- End-Level Reward: Successful completion rewards the player with Boss currency and progression to the next galaxy.

- Flow:

- Player enters the boss level.
- Player engages in combat with the boss enemy.
- Upon defeating the boss, the player is rewarded with Boss currency and progression to the next galaxy.
- The player then begins a new run in the next galaxy, starting from the base of the new reversed tree.

Ship Selection, Weapons, and Dash Mechanics in Project Pathfinder: New Home

Ship Selection

At the start of each run, the player is first presented with a variety of unique ships to select from. Each ship has its distinct traits that affect its performance and gameplay style.

Key Components:

- Ship Selection Menu: A user interface that displays the available ships along with their respective traits.
- Distinct Traits: Each ship has unique traits that define its gameplay style. These traits affect the ship's performance, such as speed, durability, and firepower.

Flow:

1. At the start of each run, the player is presented with the Ship Selection Menu.
2. The player browses through the available ships and selects one based on its distinct traits.

List of Ships:

1. Titan-Class Cruiser: The Titan-Class Cruiser is a robust and heavily armored spaceship known for its exceptional durability. Its robust structure allows it to withstand significant damage, making it a preferable choice for those who prioritize survival.

Unique Trait: The Titan-Class Cruiser's reinforced hull increases the ship's durability, allowing it to endure more hits compared to other ships.

2. Bot-Commander Frigate: The Bot-Commander Frigate is a state-of-the-art spaceship equipped with advanced robotics technology. Upon defeating an enemy, it deploys a small guided bot that emerges from the defeated enemy, tracking and targeting the closest enemy.

Unique Trait: Each enemy kill triggers the release of a guided bot from the defeated enemy. The bot tracks the nearest enemy, dealing additional damage.

3. Arcanum Explorer: The Arcanum Explorer is a highly versatile spaceship, specifically designed for treasure hunters. Its exceptional storage capacity allows it to roll more than the standard three pieces of equipment after each level.

Unique Trait: The Arcanum Explorer's extensive cargo hold allows it to roll an additional piece of equipment (4 in total) after each level.

4. Stormbringer Battleship: The Stormbringer Battleship is a formidable spaceship fitted with advanced energy manipulation technology. Each bullet fired from the Stormbringer has a chance to trigger a devastating lightning attack from above.

Unique Trait: The Stormbringer Battleship's lightning core provides a chance for each bullet to trigger a lightning attack from above, damaging nearby enemies.

5. Vampiric Destroyer: The Vampiric Destroyer is a unique spaceship equipped with life-draining technology. However, its vampiric capabilities come at a cost - the ship takes more damage from enemy attacks. Each enemy defeated restores a portion of the ship's health.

Unique Trait: The Vampiric Destroyer's energy-siphon system restores a portion of the ship's health with each enemy defeated. However, due to its vampiric nature, the ship takes additional damage from enemy attacks.

Primary and Secondary Weapons Selection

Once a ship has been selected, the player is then directed to choose one primary weapon and one secondary weapon from a selection.

Key Components:

- Weapons Selection Menu: A user interface that displays the available primary and secondary weapons and their respective attributes.

- Primary Weapon: A weapon with a high rate of fire, no cooldown, a certain magazine size, and a reload time for a new magazine. This weapon is the player's main source of damage.

- Secondary Weapon: A secondary weapon with distinct mechanics and a cooldown duration. This weapon generally deals much more damage than the primary weapon.

Flow:

1. After the ship selection, the player is directed to the Weapons Selection Menu.
2. The player selects one primary weapon and one secondary weapon based on their attributes.
3. Once both weapons have been selected, the player starts the run.

List of Primary Weapons:

1. Photon Repeater: The Photon Repeater is a highly reliable weapon, boasting a high rate of fire. Each trigger pull releases a concentrated photon projectile, making it an excellent all-around choice for any mission.

Mechanics: High rate of fire, medium magazine size, short reload time.

2. Quantum Charger: The Quantum Charger operates on a unique principle – the longer you charge, the stronger the shot. The weapon utilizes a quantum power cell that accumulates energy when the fire button is held, releasing a devastating blast when fired. The weapon requires a minimum charge before a shot can be released.

Mechanics: Charging mechanic, longer charge means higher damage, no magazine.

3. Plasma Cutter: The Plasma Cutter is a close-range, high-damage weapon. It ejects a concentrated beam of superheated plasma, acting similar to a cosmic shotgun causing substantial damage to enemies within its reach.

Mechanics: High damage, short-range, fires multiple plasma projectiles per shot, small magazine size, short reload time.

4. Neutron Bomb Launcher: The Neutron Bomb Launcher fires miniaturized neutron bombs, which detonate on impact, causing area-of-effect damage. The weapon's destructive power makes it ideal for dealing with groups of enemies.

Mechanics: Single shot, small magazine size, long reload time, bombs explode on impact causing AOE damage.

5. Bot Dispatcher: The Bot Dispatcher launches mini autonomous bots that home in on the nearest enemy, dealing damage on contact. The advanced AI of these bots ensures no shot goes wasted.

Mechanics: Medium rate of fire, bots home in on enemies, medium magazine size, short reload time.

List of Secondary Weapons:

1. Secondary Weapon 1: (Description of attributes)
2. Secondary Weapon 2: (Description of attributes)
3. Secondary Weapon 3: (Description of attributes)
4. (Continue as necessary...)

Dash Mechanics

The Dash ability allows the ship to move rapidly in a selected direction. This can be used for evasive maneuvers or positioning during combat. The Dash ability can be enhanced during a run by acquiring and equipping various types of equipment.

Key Components:

- Dash Ability: A basic ability available to all ships allowing for rapid movement in a selected direction.

- Dash Equipment: A type of equipment that enhances the Dash ability, providing additional effects during or after the Dash.

Flow:

1. The player uses the Dash ability to move rapidly in a selected direction.
2. During a run, the player can collect Dash Equipment which enhances the Dash ability.
3. Once Dash Equipment has been acquired and equipped, the enhanced Dash Ability takes effect during the game.

List of Dash Equipment:

1. Immunity During Dash: Provides the ship with immunity to damage while dashing.

2. Blink Dash: The ship teleports a short distance in the selected direction instead of dashing.

3. Decoy Drone: Leaves a decoy drone at the start of the Dash. Enemies in the radius focus on the decoy instead of the player until the decoy is destroyed.

4. Bomb Dash: Leaves a bomb at the start of the Dash. The bomb explodes after a short duration, potentially causing damage to the player.

5. Pushback Dash: Pushes back enemies in a radius around the player at the end of the Dash.

6. Bullet Dash: Fires bullets from the primary weapon in a circle around the player at the end of the Dash.

7. Shield Dash: Grants the ship a shield at the end of the Dash that can ricochet enemy bullets back.

8. Boost Dash: Enhances the ship's speed for a short duration after dashing, allowing for quick repositioning or escapes.